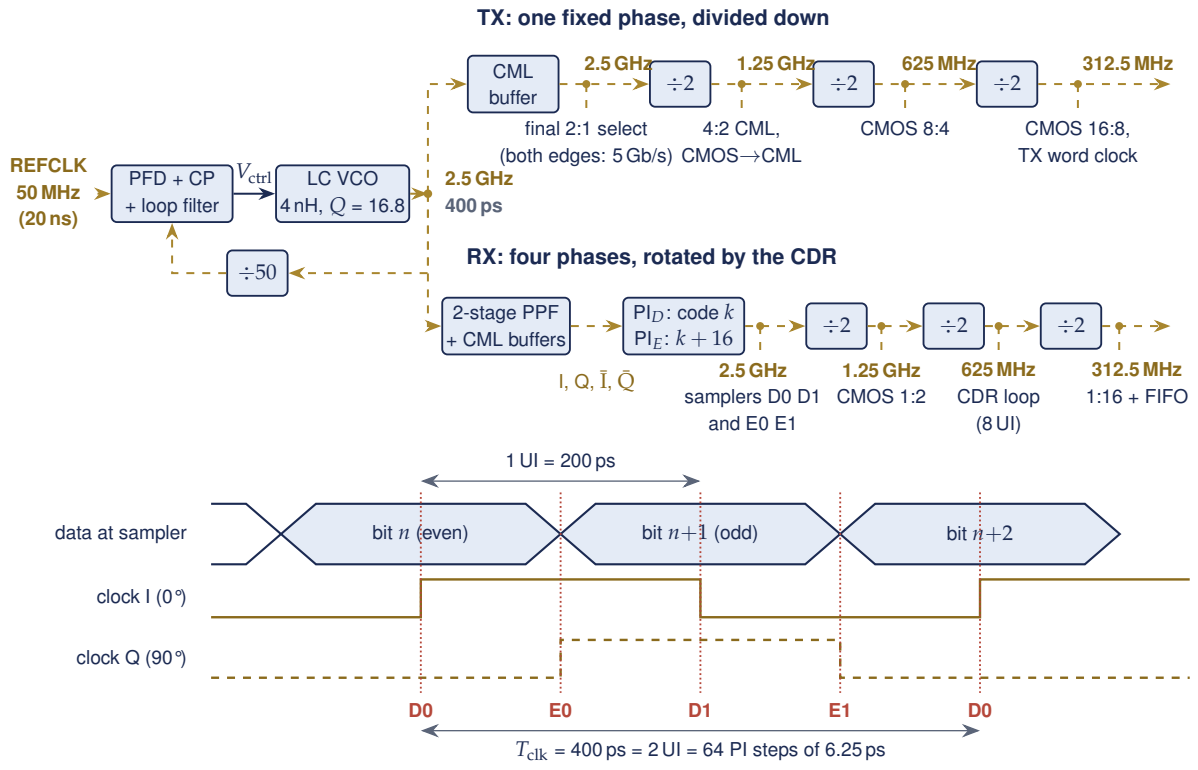

TX: one fixed phase, divided down
CML buffer
2.5 GHz
÷2
1.25 GHz
÷2
625 MHz
÷2
312.5 MHz
final 2:1 select
(both edges: 5 Gb/s)
4:2 CML,
CMOS→CML
CMOS 8:4
CMOS 16:8,
TX word clock
REFCLK
50 MHz
(20 ns)
PFD + CP
+ loop filter
V_{ctrl}
LC VCO
4 nH, Q = 16.8
2.5 GHz
400 ps
÷50
RX: four phases, rotated by the CDR
2-stage PPF
+ CML buffers
PI$_D$: code k
PI$_E$: $k+16$
I, Q, Ī, Q̄
2.5 GHz
samplers D0 D1
and E0 E1
÷2
1.25 GHz
CMOS 1:2
÷2
625 MHz
CDR loop
(8 UI)
÷2
312.5 MHz
1:16 + FIFO
1 UI = 200 ps
data at sampler
bit n (even)
bit $n+1$ (odd)
bit $n+2$
clock I (0°)
clock Q (90°)
D0
E0
D1
E1
D0
T_{clk} = 400 ps = 2 UI = 64 PI steps of 6.25 ps